Gestational diabetes and pregnancy outcomes - a systematic review of the World Health Organization (WHO) and the International Association of Diabetes in Pregnancy Study Groups (IADPSG) diagnostic criteria

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Background Two criteria based on a 2 h 75 g OGTT are being used for the diagnosis of gestational diabetes (GDM), those recommended over the years by the World Health Organization (WHO), and those recently recommended by the International Association for Diabetes in Pregnancy Study Group (IADPSG), the latter generated in the HAPO study and based on pregnancy outcomes. Our aim is to systematically review the evidence for the associations between GDM (according to these criteria) and adverse outcomes. Methods We searched relevant studies in MEDLINE, EMBASE, LILACS, the Cochrane Library, CINHAL, WHO-Afro library, IMSEAR, EMCAT, IMEMR and WPRIM. We included cohort studies permitting the evaluation of GDM diagnosed by WHO and or IADPSG criteria against adverse maternal and perinatal outcomes in untreated women. Only studies with universal application of a 75 g OGTT were included. Relative risks (RRs) and their 95% confidence intervals (CI) were obtained for each study. We combined study results using a random-effects model. Inconsistency across studies was defined by an inconsistency index (I2) > 50%. Results Data were extracted from eight studies, totaling 44,829 women. Greater risk of adverse outcomes was observed for both diagnostic criteria. When using the WHO criteria, consistent associations were seen for macrosomia (RR = 1.81; 95%CI 1.47-2.22; $p < 0.001$); large for gestational age (RR = 1.53; 95%CI 1.39-1.69; $p < 0.001$); perinatal mortality (RR = 1.55; 95% CI 0.88-2.73; $p = 0.13$); preeclampsia (RR = 1.69; 95%CI 1.31-2.18; $p < 0.001$); and cesarean delivery (RR = 1.37;95%CI 1.24-1.51; $p < 0.001$). Less data were available for the IADPSG criteria, and associations were inconsistent across studies (I2 ≥ 73%). Magnitudes of RRs and their 95%CIs were 1.73 (1.28-2.35; $p = 0.001$) for large for gestational age; 1.71 (1.38-2.13; $p < 0.001$) for preeclampsia; and 1.23 (1.01-1.51; $p = 0.04$) for cesarean delivery. Excluding either the HAPO or the EBDG studies minimally altered these associations, but the RRs seen for the IADPSG criteria were reduced after excluding HAPO. Conclusions The WHO and the IADPSG criteria for GDM identified women at a small increased risk for adverse pregnancy outcomes. Associations were of similar magnitude for both criteria. However, high inconsistency was seen for those with the IADPSG criteria. Full evaluation of the latter in settings other than HAPO requires additional studies.